# Perplexity-Guided Sampling for Efficient RL-based LLM Fine-Tuning

**Tingyi Jiang** [1]

## Abstract

Reinforcement learning (RL) has become a powerful paradigm for fine-tuning large language models (LLMs), yet the training curriculum—the order and selection of data—remains a critical bottleneck. Existing dynamic curriculum methods, such as Self-Evolving Curriculum (SEC), formulate cluster-level selection as a multi-armed bandit problem but resort to random sampling within clusters, missing fine-grained sample difficulty. In this paper, we first show that combining SEC with intra-cluster perplexity (PPL) ranking paradoxically degrades performance due to rigid cluster boundaries that restrict global sample access. Motivated by this finding, we propose **P**erplexity-**G**uided **S**ampling (PGS), a lightweight curriculum that entirely removes the bandit scheduler and instead globally ranks the entire dataset by PPL, training on the most uncertain samples at each step. PGS consistently outperforms both random sampling and SEC on the Countdown arithmetic reasoning task. On Countdown experiments, PGS achieves absolute gains of up to 4.2% and 3.7% over SEC, respectively. These results demonstrate that direct sample-level uncertainty ranking is a simpler, more effective alternative to cluster-based bandit scheduling for RL-based LLM fine-tuning.

## 1. Introduction

Reinforcement Learning (RL) methods have reshaped Large Language Model (LLM) fine-tuning, yielding notable breakthroughs with models such as OpenAI-o1 (Jaech et al., 2024) and DeepSeek-R1 (Guo et al., 2025). RL-based fine-tuning paradigms significantly advance complex reasoning capabilities in LLMs, often outperforming traditional Supervised Fine-Tuning (SFT) on complex tasks (Zhang et al., 2025).

While standard training pipelines typically employ random data sampling, recent evidence suggests that the training curriculum—the order and selection of data—is the critical bottleneck constraining fine-tuning effectiveness. Most existing Curriculum Learning (CL) strategies rely on static, predefined curricula (Zhang et al., 2025), where the sampling order is fixed prior to training and fails to adapt to the model's evolving capabilities. This rigidity leads to a difficulty shift (Zhang et al., 2025), misaligning the curriculum with the model's current learning stage and limiting final performance (Pang et al., 2026).

The Self-Evolving Curriculum (SEC) (Chen et al., 2025) addresses this by formulating curriculum learning as a dynamic Multi-Armed Bandit (MAB) problem. By adaptively selecting data clusters based on real-time learning signals, SEC demonstrates the value of dynamic scheduling. However, a critical limitation emerges when attempting to refine SEC at the sample level: while SEC effectively selects high-potential clusters, integrating perplexity-based sampling within these clusters often imposes rigid boundaries that restrict the model from accessing globally optimal training samples. This cluster-level constraint, combined with the computational overhead of maintaining Q-values, diminishes the potential gains of fine-grained selection.

Through experimentation, we empirically verify this trade-off. While SEC alone provides benefits over random sampling, its combination with intra-cluster PPL often underperforms. Based on this finding, we propose **P**erplexity-**G**uided **S**ampling (PGS). While SEC offers adaptability, we find its cluster-level constraints and computational overhead unnecessary. Unlike SEC, PGS completely removes the MAB mechanism and its associated computational costs. Instead, it directly sorts the entire dataset by perplexity (PPL), training exclusively on the most uncertain samples. This approach is remarkably lightweight yet consistently outperforms both standalone SEC and its hybrid variants by eliminating restrictive cluster boundaries.

Our work makes the following contributions:

- We demonstrate that while SEC provides dynamic scheduling benefits, its cluster-level constraints can hinder sample-level optimization when combined with PPL.
- We propose PGS, which eliminates the computational overhead of MAB scheduling (Q-value maintenance and Boltzmann sampling) while achieving superior performance.
- We show that PGS outperforms SEC and random base-

lines on both in-distribution and out-of-distribution tests, proving that direct uncertainty sampling is more effective than hybrid cluster-based approaches.

## 2. Related Works

**RL fine-tuning for LLMs.** Reinforcement Learning has become the de facto standard for enhancing reasoning capabilities in LLMs. Leveraging the Markov Decision Process (MDP) framework, sparse reward signals are typically assigned episodically after the full generation of a reasoning chain. RL fine-tuning methods, such as RLHF (Christiano et al., 2017), have demonstrated success across various domains. Policy-gradient methods, including on-policy algorithms like PPO (Schulman et al., 2017) and off-policy alternatives like DPO (Rafailov et al., 2023), are widely adopted. Algorithms such as GRPO (Shao et al., 2024) and RLOO (Ahmadian et al., 2024), have demonstrated significant gains in math and coding domains.

However, the efficacy of these methods is not solely determined by the algorithm itself but is critically constrained by the training curriculum.

**Curriculum Learning.** Curriculum Learning (CL) is a training strategy that organizes samples by increasing complexity, typically following an easy-to-hard pattern (Kim & Lee, 2024). It is widely leveraged in LLM fine-tuning to optimize training efficiency and improve final performance. Conventional CL strategies typically rely on static curricula, where the trajectory of increasing complexity is predefined offline by heuristics (Lee et al., 2024). These methods suffer from a difficulty shift, as the fixed curriculum becomes misaligned with the model's evolving capabilities.

To address this, dynamic curricula have emerged, focusing on adjusting data difficulty aligning with the model's current state. (Zhang et al., 2025) use real-time pass rates from checkpoints as a proxy for complexity, which can be time-consuming and computationally expensive. The Self-Evolving Curriculum (SEC) (Chen et al., 2025) formulates curriculum adjustment as a non-stationary Multi-Armed Bandit (MAB) problem, dynamically selecting data clusters based on the absolute advantage. While SEC effectively determines which category of data to learn, it defaults to random sampling within the selected cluster.

**Uncertainty Driven Curriculum Selection.** Recent studies highlight the importance of uncertainty in maintaining exploration. (Cui et al., 2025) demonstrate that preventing policy entropy collapse is crucial for breaking performance ceilings. (Li et al., 2026) note that high entropy in reasoning corresponds to greater sample diversity, making model training more robust. EDCO (Pang et al., 2026) utilizes total inference entropy to prioritize uncertain samples, which may be sensitive to sequence length. As a key metric reflecting generation uncertainty in LLM inference, Perplexity (PPL) is utilized in our method as a mathematically equivalent proxy that normalizes for length, providing a stable uncertainty signal.

## 3. Methods

### 3.1. Preliminary: SEC

The Self-Evolving Curriculum (SEC) (Chen et al., 2025) formulates curriculum learning as a Non-stationary Multi-Armed Bandit (MAB) problem, adapting sampling distribution based on real-time learning progress.

The training dataset $\mathcal{D}$ is partitioned into $N$ categories $C = \{c_1, ..., c_N\}$. At step $t$, SEC selects a category $c$ and samples problems uniformly from it for RL updates. The reward $r_t(c)$ for the MAB is the batch-wise average of the absolute advantage:

$$r_t(c) = \mathbb{E}_{(s_t,a_t)\sim\pi_\theta(x_i),x_i\sim c}\left[|\hat{A}_t|\right] \tag{1}$$

where $\hat{A}_t$ is computed via GRPO. SEC maintains a Q-value $Q_t(c)$ for each category, updated via Temporal Difference (TD(0)):

$$Q_{t+1}(c) = \alpha \cdot r_t(c) + (1-\alpha)\cdot Q_t(c) \tag{2}$$

Categories are sampled according to a Boltzmann distribution $p(c) \propto e^{Q_t(c)/\tau}$.

While SEC effectively selects *which category* to learn from, it defaults to uniform random sampling within the selected cluster. This overlooks fine-grained sample difficulty, burying high-advantage samples among trivial ones and leading to inefficient use of rollouts.

### 3.2. Perplexity as a Length-Normalized Uncertainty Proxy

Recent findings (Cui et al., 2025) establish a strong negative correlation between inference entropy ($\mathcal{H}$) and model performance ($R = -a\exp(\mathcal{H}) + b$). This implies that maintaining high uncertainty is crucial for preventing entropy collapse (leading to model's over-confidence at an early stage) and sustaining exploration. Inspired by EDCO (Pang et al., 2026), while it utilizes inference entropy, we adopt Perplexity (PPL), a widely applied metric of uncentainty in LLMs, as a mathematically equivalent proxy.

#### 3.2.1. Entropy or PPL

Formally, given a sequence $y$ and prompt $x$, PPL is defined as:

$$\text{PPL}(y|x) = \exp\left(-\frac{1}{L}\sum_{t=1}^{L}\log p_\theta(y_t|y_{<t},x)\right) \tag{3}$$

While as in EDCO, inference entropy is defined as:

$$H(y|x) = -\sum_{t=1}^{L} \log p_\theta(y_t|y_{<t}, x) \tag{4}$$

Although both metrics effectively identify high-uncertainty samples, PPL offers a distinct advantage in the context of SEC's intra-cluster sampling. Inference entropy is inherently biased by sequence length; longer generations accumulate higher absolute entropy values regardless of content difficulty.

Since PPL $= e^{H/L}$, it preserves the monotonicity of entropy $H$ while normalizing the uncertainty signal by the sequence length $L$. This normalization is critical for curriculum learning, as it ensures the curriculum focuses on per-token uncertainty without getting misled by the length of sequence. Its property aligns perfectly with our goal, to select samples that challenge the model's current capability boundary without being confounded by superficial features like answer length.

#### 3.2.2. Quick-Answer Prompting (QAP)

To estimate PPL efficiently without incurring the computational cost of full-sequence generation, we adopt Quick-Answer Prompting (QAP) inspired by EDCO.

We modify the input prompt $x$ to $x'$ using the QAP instruction: *"Answer concisely within three reasoning steps."* Following the QAP, we approximate PPL with response from quick prompts, which estimates the full-sequence PPL using only the first few tokens of the output, namely as Prefix PPL. The Prefix PPL Approximation is defined as:

$$\text{PPL} \approx \exp\left(-\frac{1}{L}\sum_{t=1}^{L} \log p_{\theta_k}(y_t|y_{<t}, p_{\text{quick}}, x)\right) \tag{5}$$

In this way, we guarantee both efficiency and feasibility of PPL computation.

### 3.3. Perplexity-Guided Sampling (PGS)

Based on our empirical observations, we find that the MAB mechanism in SEC does not provide significant benefits for the Countdown task. Instead, we discover that directly leveraging perplexity as the sole curriculum signal yields superior results. Therefore, we propose **P**erplexity-**G**uided **S**ampling (PGS), which completely removes the MAB scheduler and focuses on sample-level re-ranking.

The workflow of PGS is outlined in Algorithm 1. At each step $t$, we directly compute or update the PPL for samples in the dataset.

1. **Periodic PPL Update:** Every $I$ steps, we sample a subset of the dataset and compute the Prefix PPL (Eq. 5) for each sample using the current policy $\pi_\theta$ and QAP.
2. **Global Sorting:** The entire dataset $\mathcal{D}$ is sorted by PPL$[x_i]$ in descending order. This ensures that the most uncertain (and likely most informative) samples are always at the front of the queue.
3. **Batch Construction:** We simply take the first $B$ samples from the sorted list to form the training batch $\mathcal{B}_t$.

**Algorithm 1** Perplexity-Guided Sampling (PGS)

**Input:** Training set $\mathcal{D}$ partitioned into categories $C = \{c_1, ..., c_N\}$, LLM policy $\pi_\theta$, Learning rate $\alpha$, Temperature $\tau$, Batch size $B$, Total steps $T$, Prefix length $L$, Update interval $I$
**Output:** Fine-tuned LLM $\pi_\theta$
Initialize PPL$[x_i] \leftarrow 0$ for all samples in $\mathcal{D}$
**for** $t \leftarrow 0$ **to** $T-1$ **do**
  $\mathcal{B}_t \leftarrow \emptyset$
  **if** $t \bmod I = 0$ **then**
    **for** each sample $x_i$ in $\mathcal{D}$ **do**
      $x_i' \leftarrow$ QAP$(x_i)$
      Generate prefix $\tilde{y}_i \sim \pi_\theta(\cdot|x_i')$
      PPL$[x_i] \leftarrow$ Prefix PPL Approximation (Eq. 5)
    **end for**
    Sort $\mathcal{D}$ by PPL$[x_i]$ in descending order
  **end if**
  Select top $B$ samples from $\mathcal{D}$ to form $\mathcal{B}_t$
  Run $\pi_\theta$ on each $x \in \mathcal{B}_t$ to generate rollouts and compute rewards $r$
  Estimate advantages $\hat{A}$ and update $\pi_\theta$ with RL algorithm
**end for**
**Return:** Fine-tuned LLM $\pi_\theta$

**Why not combine with SEC?** We also experimented with combining SEC and PPL (i.e., using SEC to select a cluster, then sorting within that cluster by PPL). However, this approach performed worse than pure PGS. Details will be discussed in Sec. 5

## 4. Experiments

### 4.1. Setup

**Model and Dataset.** We conduct experiments on Qwen2.5-Math-1.5B model. We evaluate our method on tasks of the Countdown dataset, where given $N$ integers, use each exactly once with operators $\{+, -, \times, \div\}$ to reach a target value. The training set contains 30,000 samples uniformly divided into three difficulty levels: **i. Easy** (3 integers given, $n$=10,000); **ii. Medium** (4 integers given, $n$=10,000); and **iii. Hard** (5 integers given, $n$=10,000). These correspond to the three arms of the SEC bandit. The

test set comprises 800 samples: 600 in-distribution (ID) samples (200 each for Easy/Medium/Hard) and 200 **Very Hard** (6 integers given) out-of-distribution (OOD) samples to evaluate generalization.

**Reward Design.** We employ a rule-based 5-level scoring function (Table 1) to provide dense feedback.

*Table 1.* Rule-based Reward

| Score | Condition |
|---|---|
| 0.0 | Format error |
| 0.1 | Correct format, but numbers don't match |
| 0.2 | Format error, but numbers match |
| 0.3 | Correct format & numbers, but wrong answer |
| 1.0 | Correct format, numbers, and answer |

**Baselines.** We compare against four strategies: **i. Random.** No curriculum, uniform random sampling. **ii. SEC Bandit.** Vanilla SEC with $\alpha$=0.5, Boltzmann sampling ($\tau$=1.0), objective = |advantage|. **iii. SEC + Intra-cluster PPL.** SEC selects the cluster, then samples are sorted by PPL within that cluster (our failed attempt). **iv. PGS (Ours).** Direct global sorting by PPL without MAB.

**Implementation Details.** All methods are trained for 140 steps using Group Relative Policy Optimization (GRPO) with 8 samples per prompt and group normalization. We use AdamW optimizer with learning rate $5\times10^{-6}$. The batch size is 32. Maximum prompt length is 1024 tokens, and maximum response length is 2048 tokens. Validation is performed every 10 steps.

We adopt Quick-Answer Prompting (QAP): *"Answer the following question concisely within three reasoning steps."* Prefix length $L$=128, perplexity update interval $I$=10, and PPL sample size $S$=10,000 (33.3% of training data) per update for efficiency.

All experiments run on a single node with one NVIDIA A800 80GB GPU.

### 4.2. Main Results

Table 2 presents the main results. PGS (Ours) achieves the best performance across all in-distribution (ID) test sets and significantly outperforms all baselines on the out-of-distribution (OOD) Very Hard set. Notably, the SEC Bandit baseline performs better than Random baseline, confirming its effectiveness of dynamic curricula over random curricula. Surprisingly, the SEC + Intra-cluster PPL variant performs worse than vanilla SEC, suggesting that the interaction between cluster selection and PPL sorting is non-trivial and can degrade performance.

*Table 2.* Main results on Countdown test sets. OOD refers to Very Hard (6 numbers).

| Method | Easy (ID) | Medium (ID) | Hard (ID) | Very Hard (OOD) |
|---|---|---|---|---|
| Random | 79.5 | 56.2 | 46.7 | 34.1 |
| SEC Bandit | 90.9 | 67.5 | 61.9 | 44.8 |
| SEC + Intra-cluster PPL | 88.0 | 66.1 | 59.8 | 38.2 |
| PGS (Ours) | **95.1** | **70.5** | **67.8** | **48.5** |

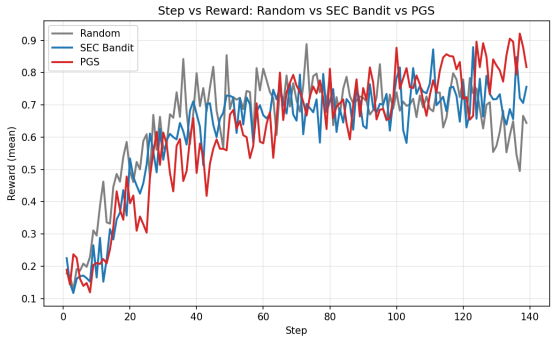



*Figure 1.* The mean reward of three methods (Random, SEC, PGS) over training steps.

We also plot the mean reward and validation results of three methods, Random, SEC and PGS, over the training, shown in Fig. 1 and Fig. 2. It can be observed that in both figures, PGS obtain the highest at the end of training. However, compare to the other two baseline methods, PGS initially lags behind during the early stages of training, but surges ahead in the later half, ultimately achieving superior final performance.

## 5. Analysis

**Uncertainty Evolution in PGS.** As shown in figures, PGS exhibits a distinct two-phase learning dynamic. Initially, PGS lags behind the SEC Bandit and Random baselines. This is attributed to the fact that at the early stage, the policy $\pi_\theta$ is nascent and lacks the foundational capability to effectively handle the massive influx of high-uncertainty samples selected by PGS. In contrast, SEC's cluster-level scheduling acts as a regularizer, providing a smoother difficulty curve that is easier for a weak policy to digest.

However, as training progresses, PGS demonstrates a stronger growth trajectory, eventually surpassing all baselines. We observe a gradual reduction in the mean uncertainty (PPL) of the training set. This indicates that the model is rapidly evolving its reasoning capabilities to resolve the previously challenging samples. By persisting in training on the uncertainty frontier, PGS prevents the model from

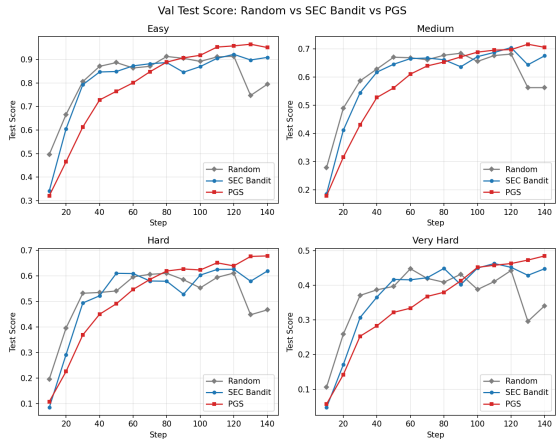



*Figure 2.* The validation result of three methods (Random, SEC, PGS) over training steps. We validation current policy every 10 steps.

plateauing on easy samples, thereby unlocking greater potential and achieving superior final performance.

**Unsatisfactory Hybrid Framework.** The hybrid SEC + Intra-cluster PPL performs worse than both standalone SEC and pure PGS. We attribute this to two key factors:

First, the hybrid framework suffers from rigid cluster confinement. Since SEC dictates that sampling must occur within a selected cluster, the model is strictly limited to the data distribution of that specific cluster. Even with PPL sorting, the model cannot access difficult samples from other clusters, creating a bottleneck that prevents it from reaching a global optimum.

Second, early-stage uncertainty is detrimental. At the beginning of training, the model's policy is too weak to handle the massive influx of high-PPL samples selected within the cluster. While pure PGS can eventually break free from this restriction and evolve by accessing diverse hard samples across clusters, the hybrid framework remains trapped. It persists in feeding the model high-uncertainty samples from a narrow domain, which stalls learning rather than promoting evolution.

**Limitations.** Due to computational resource constraints and time limit, our experiments are confined to the Countdown dataset and conducted exclusively on the Qwen2.5-Math-1.5B model. While the Countdown task provides a controlled environment for verifying algorithmic logic, a comprehensive evaluation requires broader benchmarks. Future work should validate the proposed PGS method on diverse mathematical reasoning datasets (e.g., GSM8K, MATH) to assess its generalizability across different domains and complexity levels.

Furthermore, investigating the scalability of PGS to larger foundation models remains an open avenue. With sufficient computational resources, larger models (e.g., 7B, 14B parameters) possess inherently stronger reasoning capabilities, which would allow for a more authentic verification of our method's capacity to enhance performance. In contrast, smaller models like the 1.5B used here may impose a natural performance ceiling, potentially obscuring the full potential and robustness of the PGS strategy.

## 6. Conclusion

We propose Perplexity-Guided Sampling (PGS), a lightweight curriculum for RL-based LLM fine-tuning that directly trains on globally ranked high-perplexity samples, eliminating the cluster-level Multi-Armed Bandit scheduler used in prior work like SEC. Empirically, we find that combining SEC with intra-cluster PPL sorting degrades performance due to rigid cluster boundaries. In contrast, PGS consistently outperforms random sampling, SEC, and hybrid variants on the Countdown task, achieving gains of up to 4.2% on in-distribution and 3.7% on out-of-distribution tests. Our analysis reveals a two-phase learning dynamic where PGS initially lags but ultimately surpasses baselines by persistently challenging the model at its uncertainty frontier. These results demonstrate that direct sample-level uncertainty ranking is simpler, more efficient, and more effective than cluster-based bandit scheduling. For future work, we plan to validate PGS on broader reasoning benchmarks (e.g., GSM8K, MATH) and larger LLMs to assess its generalizability.

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